Cryptography and Computational Approaches in Ensuring Data Integrity for Digital Forensic Evidence

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***Abstract*— The Digital evidence is derived from diverse sources like computers and their volatile and non-volatile storage, portable digital devices, network traffic, social networks, records and logs from internet service providers, websites, databases, and information systems on both global and local levels. This study focuses on assessing the effectiveness and appropriateness of different security approaches in safeguarding the integrity of digital evidence. The evaluation results in a series of suggestions that should be taken into account when choosing the appropriate algorithm to ensure the integrity of digital evidence in a broad sense. This research examines the cryptographic methods for safeguarding privacy in digital forensics and their classification, which is based on their support for trustworthy third parties, numerous investigations, and multi-keyword searches have been done. The Authors outline the limitations of using cryptography-based techniques in safeguarding privacy of digital forensics and propose possible remedies for these deficiencies.**

Keywords*— *digital evidence, cryptography, computational approaches, data integrity

I. Introduction

Digital forensics and investigations are being used in pandemics (for example COVID-19 contact tracking), criminal cases, civil litigations, and national security issues. Law enforcement's capacity to investigate and prosecute cybercriminals is questionable as the problem grows. Law enforcement agencies have been conducting similar investigations for years, but earlier needs assessments showed that they lack the expertise, equipment, and people to handle the volume and complexity of many of these cases [1]. In order to ensure that these digital forensics are admissible in a court of law, the Authors of this Article attempt to explain encryption, its many forms, and computational methodologies for maintaining data integrity. The present study finishes by elucidating the function of blockchain technology and the Internet of Things in data security, as well as their admissibility in legal proceedings. Additionally, the study delves into the difficulties encountered during the digital forensic investigation [2].

A. A Brief Overview of the Increasing Importance of Digital Forensics in Investigating Cyber Crimes

Early digital forensics focused on stand-alone and networked PCs. Technological advances have expanded evidence recovery to any device with a digital processor or storage. The job of digital forensics has expanded beyond investigating computer crimes like hacking to investigating all sorts of crime. With car engine management systems, satellite navigation systems, and cell phones, evidence can now include an individual's movements and activities as well as documents, images, and network activity records. Cyberspace is being used by murder, robbery, blackmail, and drug trafficking investigators to gather evidence. In the past, traditional criminal detectives ignored digital evidence because they didn't grasp its importance. This is changing, but conventional crime detectives have a long way to go before they appreciate the potential worth of digital evidence and have the resources to handle it [3].

B. The Critical Need for Ensuring Data Integrity in Digital Forensic Evidence

Each and every legal evidence should be authenticated. The plaintiff must provide relevant, authentic, and undamaged evidence. They must also demonstrate a proper chain of custody, which involves recording evidence handling. The court can ensure the evidence was handled properly to preserve its validity and integrity. In court, evidence cannot be admitted if the plaintiff cannot prove chain of possession and validity. Expert corporate forensics specialists verify the admissibility of every digital evidence they uncover [4].

C. Review of Literature

The Authors have gone through the various Literature for this Study. Some of them have been discussed here. David Mughisha in their Paper, “Role and Impact of Digital Forensics in Cyber Crime Investigation” discusses about Cyber Crime and Global Economic Growth, Reasons for conducting a Digital Forensic Investigation, Legal Aspects and What the Future holds in the field of Digital Forensics. Josiah Dytkstyra in their Book “Cloud Technology” investigates how to obtain Forensic Evidence from Cloud Computing using the Legal Process by surveying the existing statistics and recent cases applicable to Cloud Forensics. This Book discusses how Defense Attorneys might be able to challenge Cloud-Derived Evidence in the Court. Rahman in “Securing Digital Forensic Investigation Data in the Cloud: Review, Taxonomy and Open Challenges” discusses about the meaning and classification of the Securing Digital Forensic Investigation Data in the Cloud. This Study not only reviews the earlier Study in this regard but also talks about the challenges associated with it. Shahzad Saleem, Oliver Popov and Rami Dahman in “Evaluation of security methods for ensuring the integrity of digital evidence" states that the Acquisition and Analysis of the Electronic Evidence is crucial for the understanding and functioning of the Digital World. It states that, to protect the integrity of Digital Evidence, a number of Security Methods are used. This work deals with the Evaluation of these Security Methods in order to understand their goodness and suitability to protect the

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integrity of the Digital Evidence. Its outcome is to evaluate and to give the set of recommendations to be considered for selecting the right algorithm to protect the integrity of Digital Evidence. Anwar Iftikar, etall. in "Digital Rorensic Investigations: Optimal Strategies and Emerging Innovations" discuss the foundational concepts of Quality Control in Digital Forensics, emphasizing the of Standardized Proceedings, Documentations and Validation Techniques. It delves into the Potential Sources of Errors and Bias that can arise during the Acquisition, Preservation, Analysis and Presentation of Digital Evidence. This Review Article offers Researchers, Practioners and Policy Makers an insight into the enhancing the Reliability and Trustworthiness of Digital Forensic Investigations.

The Present Study, however is in the furtherance of the above mentioned Literature. The Authors in the current Study talks about the Integrity of Digital Evidence. Cryptographic and Computational Approaches have been discussed and the role of their Integration in the Admissibility of such Digital Evidence has been delved into. Various Threats and Challenges and Future Trends have been discussed, and thereby Certain Guidelines have been figured out to facilitate the use and Admissibility of the Computational and Cryptographic Techniques as Evidence in the court of law.

II. THE DIGITAL FORENSIC INVESTIGATION PROCESS

A. *Stages Involved in the Digital Forensic Process*

Digital forensics recovers, investigates, and analyses electronic data for legal, criminal, and internal investigations. Electronic data may help solve cybercrime, data theft, crypto crimes, security breaches, hacking, and more. Digital forensics help solve complicated data problems. Digital forensic investigators utilise various tools and software to investigate, and they try to: Identify the source and reason of a cyberattack, to determine if a hack was committed and the duration of the hacker's access, to create a chronology of illegal actions, such as unauthorised data access or alteration, Safeguard digital evidence. Digital forensic investigations are capable of detecting and proving financial fraud, espionage, network or system breaches, data theft, and internet abuse. In civil or criminal actions, ensuring data integrity and court admissibility requires a methodical, process-driven digital forensics investigation. The identification of resources and devices engaged in the inquiry, the preservation of pertinent data, the analysis, documentation, and presentation of the evidence are among the fundamental investigative procedures in digital forensics. The processes involved in digital forensic investigation are depicted in Fig. 1.

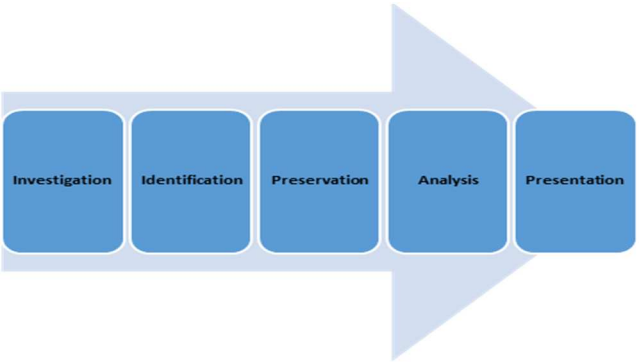

Fig.1. Process of Digital Forensics

This type of data can be utilised in court to support lawsuit. Digital forensics investigators are trained to collect and handle court-admissible evidence, which may be crucial in digital data lawsuit cases [5].

B. *Integration of Data Integrity, cryptography and Blockchain in the Digital Forensic Investigation Process*

An important development in the field of digital forensic inquiry is the combination of blockchain technology, cryptography, and data integrity. By using encryption techniques to protect sensitive data from unauthorised access or alteration, cryptography guarantees that data is safe and unchangeable. By maintaining the quality and dependability of the data throughout forensic examination, this cryptographic protection improves data integrity. These ideas are further supported by blockchain technology, which offers a decentralised, immutable ledger where data exchanges are safely documented and timestamped. Throughout the course of an investigation, forensic specialists may track and confirm the legitimacy of digital evidence by utilising blockchain technology, which preserves the evidence's integrity. Cryptography and blockchain together allow forensic investigators to improve the robustness and reliability of their investigations in today's increasingly digital environment by strengthening their methods, reducing the possibility of data manipulation, and boosting confidence in digital evidence.

III. THREATS TO DATA INTEGRITY

Data integrity assures comprehensive, consistent, accurate, and up-to-date data from collection to destruction. Data integrity failure hinders management and decision-making and raises safety hazards.Hacking, data breaches, viruses, and defects are certain security threats which are generaly faced[6]. Attacks on data integrity put digital forensic investigations at serious risk by undermining the dependability and precision of evidence analysis. The credibility of digital records and artefacts can be compromised by malicious actions such data alteration, tampering, and unauthorised access, which can impede the investigation process [7]. Furthermore, as adversaries are always looking for ways to exploit holes in digital systems and networks, the growth of sophisticated cyberattacks and malware presents significant dangers to data integrity[8]. Moreover, data integrity risks might result in data loss or corruption due to accidental mistakes, hardware malfunctions, and software bugs. Strong security measures, such as encryption, access limits, and ongoing monitoring, are needed to address these issues and protect data integrity as well as the integrity of forensic analysis results.

In legal contexts, compromising data integrity can have dire consequences since it erodes the basis of the court system's dependability and trustworthiness. The quality, consistency, and dependability of information are referred to as data integrity, and any breach in these areas may have dire legal repercussions. For fair and reasonable outcomes in judicial procedures, the accuracy of the evidence and information is crucial. Data integrity breaches have the potential to lead to incorrect verdicts, unfair trials, and harm the judicial system's reputation. Compromised data also poses a risk to the integrity of the judicial system as a whole by undermining public confidence in the legal system. Individuals or groups guilty of such compromises may be

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subject to civil and criminal penalties, including fines and imprisonment, in addition to any legal responsibilities. In addition to being required by law, ensuring data integrity is also crucial for respecting the values of justice and preserving public trust in the judicial system [9].

IV. Fundamentals And Types Of Cryptography

A. Fundamentals of Cryptography

Cryptography studies and practices secure communication with adversaries. It develops and analyses procedures to prevent hostile third parties from accessing shared information, using information security principles. Secure communication prevents an enemy from accessing two parties' messages or data. An adversary in cryptography is a malevolent entity that tries to steal data and undermine information security. Modern cryptography emphasises confidentiality, integrity, authentication, and non-repudiation.
(1) Confidentiality refers to rules and procedures that restrict information to certain individuals or locations, generally through confidentiality agreements.
(2) Data integrity involves ensuring correct and consistent data throughout its life cycle.
(3) Authentication verifies that the data claimed by the user belongs to them.
(4) Non-repudiation ensures that a person or party involved in a contract or communication cannot reject the legitimacy of their signature or message [10].

B. Types of Cryptography

1.The Symmetric Key Cryptography technique uses a single key to encrypt and decode communications for both sender and recipient. Symmetric Key Systems are quicker and simpler, but sender and receiver must securely exchange keys. The most common symmetric key cryptography systems are DES (Data Encryption Standard) and AES (Advanced Encryption Standard) .
2. Hash Functions: No key is used in this method. A fixed-length hash value is computed from plain text, making it hard to recover. Many operating systems encrypt passwords with hash algorithms.
3. Asymmetric Key Cryptography: This approach encrypts and decrypts information using two keys. Receiver private keys are used for decryption and public keys for encryption. The intended recipient can only decode it because he knows his private key, even if everyone knows the public key [11].

V. Application Of Block Chain Technology

Block-chain, widely known for crypto currency, offers an intriguing cyber security solution. Block-chain prevents data modification well. Security for IoT devices is supported. IoT stands for Internet of Things. Hacked IoT network instruments can be closed and disconnected. The blockchain is a distributed ledger. It enables machine-to-machine interactions with IoT. It uses a transaction database. Central ledgers double-check and record these transactions [12].

A. Investigating Block Chain's Potential Benefit in Protecting Data Integrity in the Field of Digital Forensics

Block-chain, widely known for crypto money, offers an innovative cyber security option. Block-chain prevents data manipulation. In dispersed environments, it protects IoT devices. Compromised IoT network devices can be closed and unplugged. A distributed ledger system is blockchain. It enables machine-to-machine communication via the Internet of Things. It uses a transaction database. These transactions are triple-checked before entering a central ledger. These ledgers are on each node. This study examines block chain and the Internet of Things. This article reviews blockchain and IoT research. The development and application issues of block-chain have been disputed. The relevance of block chain in IoT-based research for online transaction security is also examined [13].

B. Exploring How Blockchain Technology is Used to Build Tamperproof Audit Trails

Blockchain technology, which is known for being decentralised and unchangeable, offers a creative way to create audit trails that cannot be tampered with. Because blockchain is decentralised, it distributes the ledger among a network of nodes, doing away with the necessity for a central authority. Transparency is maintained and a single point of failure is avoided since every participant has a copy of the complete chain. Consensus procedures and cryptographic hash functions provide immutability, a crucial feature. Because the ledger is distributed, modifying any information after a block is added to the chain would need modifying following blocks, which is computationally impractical. Blockchain is a perfect technology for producing audit trails that are difficult to tamper with because of its inherent immutability. The blockchain creates a chronological and unchangeable history by cryptographically linking every transaction and event that is recorded. Therefore, boosting accountability, transparency, and confidence in data integrity, blockchain-based tamperproof audit trails find applications in a variety of areas, from financial transactions to supply chain management [14].

VI. Computational Approaches

A. Computational Methods: Machine Learning Algorithms for Identifying and Tracing Anomalies in Large Datasets

IoT ideas like cloud, fog, edge, and mist computing are well-known.Recently developed processing on the network edge has lowered latency, privacy, and location awareness demands while boosting privacy by not transmitting raw data to the cloud. The writers discuss fog, edge, mist, and mobile computing. Sometimes these computer paradigms are ambiguous, allowing for additional sensor and IoT device location research. Classifying and analysing edge computing architectures and platforms exposes hidden network edges. Resource-rich edge servers, heterogeneous edge nodes, and edge-cloud federation are possible. The second section of the study shows edge computing's IoT benefits via mobile gaming. Placing limited processing resources near the network edge improved latency and game responsiveness [15].

Many fields use deep learning successfully. The extended assessment found that it reduces latency, improves privacy, bandwidth efficiency, and scalability while processing data on edge devices. Deep learning is used for computer vision, NLP (Natural Language Processing), and network intrusion detection. DNNs (Deep Neural Network) are complex and resource-intensive, therefore model construction, compression, and specialised hardware must be tuned. Cyber-security IoT applications push traffic inspection and classification to edge devices for processing intelligence near the data source. Devices run pre-built categorization models,

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whereas cloud-based systems train data computationally. A cloud-assisted Extreme Learning Machine (ELM) classifier effectively calculated and analysed data in an autonomous attack detection system[10].
WSN security involves anomaly detection. We must decrease false alarms. Machine learning algorithms are often used. Training and evaluation generally require the node dataset. By employing local functions on a sample of data, an online locally weighted projection regression (OLWPR) model for WSN anomaly detection lowers computing costs. In terms of accuracy, F1 score, average RMSE, and average percentage error, OLWPR fared better than logistic regression (LR), random forest (RF), AdaBoost, decision tree (DT), support vector machine (SVM), and Artificial Neural Network (ANN) [16].
In a different anomaly detection scenario, machine learning forecasted IOT abnormalities and assaults. The LR, SVM, DT, RF, and ANN supervisory learning techniques find abnormalities in the DS2OS (Distributed Smart Space Orchestration System) dataset. Accuracy, precision, recall, F1 score, confusion matrix, and ROC (Receiver Operating Characteristic) curve area were used to compare performance. Using an OC-SVM (One Class Support Vector Machine) and an LSTM auto-encoder, imbalanced data is identified. Input data is compressed using the LSTM (Long Short Term Memory) auto-encoder using traffic patterns. Unusual attacks are identified using OC-SVM. Using the most recent dataset (InSDN) of software-defined network intrusion detection systems (IDSs), the suggested model detects malicious traffic with exceptional accuracy and little processing time.[17] Researchers have been examining abnormalities for years. Deep learning has become more and more popular recently. Research on cutting edge data mining and machine learning technology is presented at conferences and in publications. The writers talk about pre-training models, designs, open-source techniques, and the intricacy of issues. IOT device problems may arise from integrated security and insufficient resources. Increased intrusion assaults may lead to errors in the current intrusion detection systems, which are insufficient. On Bot-IoT datasets, contemporary machine learning techniques are compared for binary and multiclass classification. Data expertise is required since most edge devices, on which machine learning algorithms operate, have limited memory and processing power. A thorough model, architecture, and requirements analysis demonstrate the operation of ML algorithms on a sizable, lopsided dataset [18].

B. Computational Methods Used in Conjunction with Cryptographic Techniques to Strengthen Data Security

When combined with cryptographic approaches, computational methods are a powerful tool for enhancing data security in modern digital contexts. An essential layer of security is provided by cryptographic techniques like encryption algorithms, which convert plaintext data into incomprehensible ciphertext and make it unreadable by unauthorised parties. The basis of encryption is complemented by computational techniques, such as sophisticated algorithms and protocols, which increase the complexity and robustness of security measures [19]. An additional line of protection is added by the ease with which multi-factor authentication systems may be implemented thanks to the combination of computational techniques and encryption. In addition to strengthening data confidentiality, this cooperative effort tackles the ever-changing landscape of cybersecurity issues by providing an advanced and flexible barrier against possible intrusions [20].

VII. Future Trends And Challenges

Because of its significance in cybersecurity, researchers and practitioners have concentrated on digital forensics. Due of the complexity of gathering evidence, which usually requires many technologies, modern hacks are more sophisticated. Scholars have carried out several field surveys and assessments [21]. These studies, however, focused on the advancements in each area of digital forensics specifically. Although the worldwide viewpoint is lacking, each of these surveys aids academics and professionals in staying current with advancements in the field of digital forensics. To close this gap, we identified the main topics and issues through a qualitative assessment of all pertinent digital forensics reports. The majority of them share worries about acquiring evidence and pre-processing because of counter-analytical techniques and challenges in obtaining data from devices, the cloud, etc., even though their topics and tactics are different [22]. The developments and obstacles in the field of digital forensic investigation are illustrated in Fig. 2.

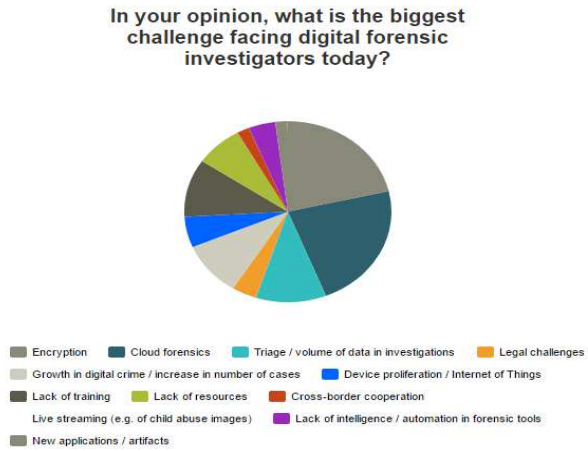

Fig.2 Future Trends and challenges in Digital forensic investigation

VIII. Law And Challengeshreats

Due to our everyday usage of technology, cyberspace digital evidence research is growing. New methods for gathering, analysing, and preserving digital evidence have emerged due to the expanding volume of digital data. Digital forensics researchers and practitioners confront several hurdles, including fast technological change, privacy and data protection problems, and a complicated legal framework for gathering and analysing digital evidence [23]. Despite these obstacles, digital evidence is vital to law enforcement and the judicial system in criminal and civil proceedings. Digital forensics must adapt to new threats and problems as technology advances. Recently, digital technology has become widespread in many aspects of society, including crime. Cybercrimes today threaten society due to rapid technological advancement. Digital evidence from computer devices is crucial to criminal investigations. It may be difficult to establish that Courts accept digital evidence [24].

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IX. General Guidelines To Facilitate The Use And Admissibility Of The Computational And Cryptograpohic Techniques As Evidence In The Court Of Law

A few broad rules need to be adhered to in order to guarantee the use and admissibility of computational and cryptographic methods as evidence in a court of law. First and foremost, thorough documenting of the processes and procedures used during data processing and analysis is crucial for ensuring transparency and repeatability in the use of these approaches. Furthermore, in order to ensure the correctness and dependability of the computational algorithms and cryptographic protocols utilised, they should undergo validation and peer review by acknowledged authorities in the area. Furthermore, it is essential to appropriately document evidence management protocols and any modifications made to the data in order to preserve the validity and integrity of digital evidence along the chain of custody. In addition, it is crucial to follow legal guidelines and rules controlling the use of digital evidence, such as the US Federal Rules of Evidence, in order to guarantee its admissibility in court. The acceptance of the evidence in a court of law can be facilitated by providing judges and juries with a clear and concise presentation of the findings, as well as expert testimony that explains the underlying computational and cryptographic techniques used.

X. Conclusions

Given our daily use of technology, digital forensics is essential to fighting cybercrimes and other legal challenges. Technology allows digital forensics to retrieve evidence from various devices besides computers. This study revealed that digital evidence integrity, which required good security, made digital forensics valuable. Court processes depend on digital forensic evidence data legality and dependability. Court mistrust and wrongful convictions might come from data integrity breaches. Cryptography and computation safeguard digital forensics data. Cryptography ensures privacy, integrity, authentication, and non-repudiation. Machine learning anomaly detection, pattern recognition, and key management improve security. Decentralised, unchangeable blockchain technology is being studied for data integrity and audit trails by digital forensics specialists. Blockchain and IoT provide exciting security alternatives. Future digital forensics issues may require interdisciplinary teams and global viewpoints. The study's qualitative examination showed comparable concerns in evidence collection, pre-processing, procedural preparation, reporting, and ethics. To grow, digital forensics must confront cyberthreats. Legality of digital evidence matters. Digital forensics faces complex legal, privacy, and data protection issues. Digital evidence is crucial to criminal and civil proceedings and to ensure its admissibility in the Court of Law all the above mentioned components of Data Integrity are indispensable.

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